

$$C_{\phi e}^{(12)} i_1 i_2 \rightarrow$$

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(-\frac{1}{3} \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} LF_{2,2,0} [m_d^r, m_q^p] \delta_{i_1 i_2} + \frac{1}{6} \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} LF_{3,2,-1} [m_d^r, m_q^p] \delta_{i_1 i_2} - \right. \\
& \frac{1}{3} \tilde{\mu} g_1^2 \overline{y_e^{pr}} a_e^{pr} LF_{2,2,0} [m_e^r, m_l^p] \delta_{i_1 i_2} + \frac{1}{6} \tilde{\mu} g_1^2 \overline{y_e^{pr}} a_e^{pr} LF_{3,2,-1} [m_e^r, m_l^p] \delta_{i_1 i_2} + \\
& \frac{1}{3} \tilde{\mu} g_1^2 \overline{y_e^{pr}} a_e^{pr} LF_{3,1,0} [m_l^p, m_e^r] \delta_{i_1 i_2} + \frac{1}{3} \tilde{\mu} g_1^2 \overline{y_e^{pr}} a_e^{pr} LF_{3,2,-1} [m_l^p, m_e^r] \delta_{i_1 i_2} - \\
& \frac{3}{4} \tilde{\mu} g_1^2 \overline{y_e^{pr}} a_e^{pr} LF_{4,1,-1} [m_l^p, m_e^r] \delta_{i_1 i_2} + \frac{1}{3} \tilde{\mu} g_1^2 \overline{y_e^{pr}} a_e^{pr} LF_{5,1,-2} [m_l^p, m_e^r] \delta_{i_1 i_2} + \\
& \frac{1}{3} \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} LF_{3,1,0} [m_q^p, m_d^r] \delta_{i_1 i_2} + \frac{1}{3} \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} LF_{3,2,-1} [m_q^p, m_d^r] \delta_{i_1 i_2} - \\
& \frac{5}{4} \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} LF_{4,1,-1} [m_q^p, m_d^r] \delta_{i_1 i_2} + \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} LF_{5,1,-2} [m_q^p, m_d^r] \delta_{i_1 i_2} + \\
& \frac{1}{6} \tilde{\mu} g_1^2 \overline{y_u^{pr}} a_u^{pr} LF_{2,2,0} [m_q^p, m_u^r] \delta_{i_1 i_2} - \frac{1}{12} \tilde{\mu} g_1^2 \overline{y_u^{pr}} a_u^{pr} LF_{3,2,-1} [m_q^p, m_u^r] \delta_{i_1 i_2} - \\
& \frac{1}{6} \tilde{\mu} g_1^2 \overline{y_u^{pr}} a_u^{pr} LF_{3,1,0} [m_u^r, m_q^p] \delta_{i_1 i_2} - \frac{1}{6} \tilde{\mu} g_1^2 \overline{y_u^{pr}} a_u^{pr} LF_{3,2,-1} [m_u^r, m_q^p] \delta_{i_1 i_2} - \\
& \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_u^{pr}} a_u^{pr} LF_{4,1,-1} [m_u^r, m_q^p] \delta_{i_1 i_2} + \tilde{\mu} g_1^2 \overline{y_u^{pr}} a_u^{pr} LF_{5,1,-2} [m_u^r, m_q^p] \delta_{i_1 i_2} - \\
& \frac{1}{4} m_1 \tilde{\mu} g_1^4 LF_{4,1,-1} [\tilde{\mu}, m_1] \delta_{i_1 i_2} + \frac{1}{3} m_1 \tilde{\mu} g_1^4 LF_{5,1,-2} [\tilde{\mu}, m_1] \delta_{i_1 i_2} - \\
& \frac{3}{4} m_2 \tilde{\mu} g_1^2 g_2^2 LF_{4,1,-1} [\tilde{\mu}, m_2] \delta_{i_1 i_2} + m_2 \tilde{\mu} g_1^2 g_2^2 LF_{5,1,-2} [\tilde{\mu}, m_2] \delta_{i_1 i_2} + \\
& \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{2,1,1,0} [m_1, m_e^{i1}, m_l^p] - \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{2,2,1,-1} [m_1, m_e^{i1}, m_l^p] - \\
& \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{3,1,1,-1} [m_1, m_e^{i1}, m_l^p] + \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{2,2,1,-1} [m_1, m_l^p, m_e^{i1}] + \\
& \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{2,1,1,0} [m_1, m_l^p, \tilde{\mu}] - \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{3,1,1,-1} [m_1, m_l^p, \tilde{\mu}] + \\
& \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{2,2,1,-1} [m_1, \tilde{\mu}, m_e^{i1}] - \frac{3}{4} m_2 \tilde{\mu} g_2^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{2,1,1,0} [m_2, m_l^p, \tilde{\mu}] + \\
& \frac{3}{4} m_2 \tilde{\mu} g_2^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{3,1,1,-1} [m_2, m_l^p, \tilde{\mu}] + \frac{3}{4} m_2 \tilde{\mu} g_2^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{2,2,1,-1} [m_2, \tilde{\mu}, m_l^p] - \\
& \frac{1}{2} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{2,1,1,0} [\tilde{\mu}, m_1, m_e^{i1}] + \frac{1}{2} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{3,1,1,-1} [\tilde{\mu}, m_1, m_e^{i1}] + \\
& \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{1,1,1,1,0} [m_1, m_e^{i1}, m_l^p, \tilde{\mu}] + \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} \\
& LF_{2,1,1,1,-1} [m_1, m_e^{i1}, m_l^p, \tilde{\mu}] + \frac{1}{4} \tilde{\mu} g_1^2 \overline{y_e^{pi1}} a_e^{pi2} LF_{1,1,1,1,0} [m_1, m_e^{i2}, m_l^p, \tilde{\mu}] - \\
& \frac{1}{8} \tilde{\mu} g_1^2 \overline{y_e^{pi1}} a_e^{pi2} LF_{2,1,1,1,-1} [m_1, m_e^{i2}, m_l^p, \tilde{\mu}] - \frac{1}{4} \tilde{\mu} \overline{y_e^{pi1}} \overline{y_e^{rs}} y_e^{ri2} a_e^{ps} \\
& LF_{2,1,1,1,-1} [m_e^s, m_l^p, m_l^r, \tilde{\mu}] + \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{2,1,1,1,-1} [m_e^{i1}, m_1, m_l^p, \tilde{\mu}] - \\
& \frac{1}{8} \tilde{\mu} g_1^2 \overline{y_e^{pi1}} a_e^{pi2} LF_{2,1,1,1,-1} [m_e^{i2}, m_1, m_l^p, \tilde{\mu}] + \frac{1}{2} \tilde{\mu} g_1^2 \overline{y_e^{pi1}} a_e^{pi2} \\
& LF_{2,1,1,1,-1} [m_l^p, m_1, m_e^{i1}, m_e^{i2}] - \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} LF_{2,1,1,1,-1} [m_l^p, m_1, m_e^{i1}, \tilde{\mu}] + \\
& \frac{1}{8} \tilde{\mu} g_1^2 \overline{y_e^{pi1}} a_e^{pi2} LF_{2,1,1,1,-1} [m_l^p, m_1, m_e^{i2}, \tilde{\mu}] - \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{pi1}} y_e^{pi2} \\
& LF_{2,1,1,1,-1} [\tilde{\mu}, m_1, m_e^{i1}, m_l^p] + \frac{1}{8} \tilde{\mu} g_1^2 \overline{y_e^{pi1}} a_e^{pi2} LF_{2,1,1,1,-1} [\tilde{\mu}, m_1, m_e^{i2}, m_l^p] \Big)
\end{aligned}$$